

## 15. IMPLEMENT IRIS FLOWER CLASSIFICATION USING NAIVE BAYES CLASSIFIER

### 1. Iris Dataset 1

Dataset: iris1.csv

| Sepal Length | Sepal Width | Petal Length | Petal Width | Species    |
|--------------|-------------|--------------|-------------|------------|
| 5.1          | 3.5         | 1.4          | 0.2         | Setosa     |
| 4.9          | 3.0         | 1.4          | 0.2         | Setosa     |
| 6.0          | 2.9         | 4.5          | 1.5         | Versicolor |
| 5.9          | 3.0         | 4.2          | 1.5         | Versicolor |
| 6.5          | 3.0         | 5.2          | 2.0         | Virginica  |
| 6.7          | 3.1         | 5.6          | 2.4         | Virginica  |

### PROGRAM

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB
df = pd.read_csv("iris1.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = GaussianNB()
model.fit(X, y)
sample = [[6.4, 3.0, 5.1, 1.9]]
print("Predicted Flower:", model.predict(sample)[0])
```

### OUTPUT

Predicted Flower: Virginica

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### 2. Iris Dataset 2

### Dataset: iris2.csv

| Sepal Length | Sepal Width | Petal Length | Petal Width | Species    |
|--------------|-------------|--------------|-------------|------------|
| 5.0          | 3.4         | 1.5          | 0.2         | Setosa     |
| 5.2          | 3.5         | 1.5          | 0.2         | Setosa     |
| 5.8          | 2.7         | 4.1          | 1.0         | Versicolor |
| 6.1          | 2.8         | 4.0          | 1.3         | Versicolor |
| 6.3          | 2.9         | 5.6          | 1.8         | Virginica  |
| 6.7          | 3.0         | 5.2          | 2.3         | Virginica  |

### PROGRAM

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB
df = pd.read_csv("iris2.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = GaussianNB()
model.fit(X, y)

sample = [[5.1, 3.4, 1.5, 0.2]]
print("Predicted Flower:", model.predict(sample)[0])
```

### OUTPUT

Predicted Flower: Setosa

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### 3. Iris Dataset 3

#### Dataset: iris3.csv

| Sepal Length | Sepal Width | Petal Length | Petal Width | Species |
|--------------|-------------|--------------|-------------|---------|
| 5.4          | 3.7         | 1.5          | 0.2         | Setosa  |

|     |     |     |     |            |
|-----|-----|-----|-----|------------|
| 4.8 | 3.4 | 1.6 | 0.2 | Setosa     |
| 6.0 | 2.7 | 5.1 | 1.6 | Versicolor |
| 5.7 | 2.8 | 4.5 | 1.3 | Versicolor |
| 6.2 | 3.4 | 5.4 | 2.3 | Virginica  |
| 6.4 | 3.2 | 5.3 | 2.3 | Virginica  |

## PROGRAM

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB
df = pd.read_csv("iris3.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = GaussianNB()
model.fit(X, y)
sample = [[5.9, 2.8, 4.7, 1.2]]
print("Predicted Flower:", model.predict(sample)[0])
```

## OUTPUT

Predicted Flower: Versicolor

## 4. Iris Dataset 4

**Dataset: iris4.csv**

| Sepal Length | Sepal Width | Petal Length | Petal Width | Species    |
|--------------|-------------|--------------|-------------|------------|
| 5.1          | 3.8         | 1.5          | 0.3         | Setosa     |
| 4.7          | 3.2         | 1.3          | 0.2         | Setosa     |
| 6.2          | 2.9         | 4.3          | 1.3         | Versicolor |
| 5.6          | 2.9         | 3.6          | 1.3         | Versicolor |
| 6.8          | 3.2         | 5.9          | 2.3         | Virginica  |

|     |     |     |     |           |
|-----|-----|-----|-----|-----------|
| 7.1 | 3.0 | 5.9 | 2.1 | Virginica |
|-----|-----|-----|-----|-----------|

## PROGRAM

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB

df= pd.read_csv("iris4.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = GaussianNB()
model.fit(X, y)
sample = [[6.9, 3.1, 5.8, 2.2]]
print("Predicted Flower:", model.predict(sample)[0])
```

## OUTPUT

Predicted Flower: Virginica

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## 5. Iris Dataset 5

**Dataset: iris5.csv**

| Sepal Length | Sepal Width | Petal Length | Petal Width | Species    |
|--------------|-------------|--------------|-------------|------------|
| 5.0          | 3.5         | 1.3          | 0.3         | Setosa     |
| 5.3          | 3.7         | 1.5          | 0.2         | Setosa     |
| 5.9          | 3.0         | 4.2          | 1.5         | Versicolor |
| 6.0          | 2.9         | 4.5          | 1.5         | Versicolor |
| 6.5          | 3.2         | 5.1          | 2.0         | Virginica  |
| 6.7          | 3.3         | 5.7          | 2.1         | Virginica  |

## **PROGRAM**

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB
df = pd.read_csv("iris5.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]

model = GaussianNB()
model.fit(X, y)
sample = [[5.8, 3.0, 4.1, 1.4]]
print("Predicted Flower:", model.predict(sample)[0])
```

## **OUTPUT**

Predicted Flower: Versicolor